

Class 6: Financial Statement Analysis

PRACTICE QUESTIONS: SOLUTIONS

These practice questions are designed to help improve your understanding of the course material. The questions referenced in this document can be found at the **end of each chapter in your textbook**.

Use the **Table of Contents** below to identify the list of questions and the **suggested time** for completing each one. When using this resource:

- ☐ **Do the exercises soon after the lecture** and ideally **before tutorials**. Studies on learning show that practicing material shortly after it's introduced improves understanding and retention. Spacing out your practice and reviewing regularly also strengthens long-term memory.
- ☐ **Attempt the exercises first** without looking at the solutions or debriefing notes. This will help you simulate real exam conditions and identify areas where you may need further practice.
- ☐ Be **conscious of the suggested time** for each question. Staying aware of time management is a valuable skill for exams.
- ☐ After completing the exercises, use the solutions provided to **debrief your work**. Compare your answers, understand any mistakes, and learn from them to solidify your understanding of the concepts.

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IMPORTANT: Additional DuPont practice case (Costco) is posted on Quercus. Please make sure to complete it.

CH. 12 | ANALYSING AND INTERPRETING FINANCIAL STATEMENTS

M12-23 (~ 10 min)

BMW Group (€ millions)

- a. BMW's current ratio for the two years presented is as follows:

2019 current ratio: $\text{€ } 90,630 / \text{€ } 82,625 = 1.097$

2018 current ratio: $= 1.178$

BMW's current ratio of 1.097 and 1.178 closely approximates the industry average of 1.19 (we use the "telecommunications" to proxy for the "automotive" industry). This indicates that its current assets are just sufficient to cover its current liabilities but slightly below the industry standard. We might want to know, however, whether BMW's current assets are concentrated in cash or relatively illiquid inventories, as well as the maturity schedule of its current liabilities. We would also like to know the median current ratio for the industry. This would help place BMW's numbers in perspective.

- b. BMW's times interest earned ratio for the two years is as follows:

2019 times interest earned $= \text{€ } 2,276 / \text{€ } 499 = 4.56$

2018 times interest earned $= 10.77$

2019 debt-to-equity $= \text{€ } 168,127 / \text{€ } 59,324 = 2.83$

2018 debt-to-equity $= 2.62$

BMW has a good ability to meet its debt obligations. A healthy TIE of 4.56 indicates that BMW's ability to cover interest expense with pre-tax income is satisfactory. However, the ratio is not as strong as that in 2018, due to lower EBIT and increased interest expense in 2019. D/E ratio (2.83) is slightly aggressive in financing with debts for its growth. This also indicates that BMW could have more volatile earnings as a result of the financial leverage.

M12-24 (~ 20 min)

a.

\$ millions	Asset Turnover
Procter & Gamble	$\$83,680/\$135,299 = 0.62$
McDonald's	$\$27,006/\$32,483 = 0.83$
Valero Energy	$\$125,987/\$40,202 = 3.13$

b.

\$ millions	ART
Procter & Gamble	$\$83,680/\$6,172 = 13.56$
McDonald's	$\$27,006/\$1,257 = 21.48$
Valero Energy	$\$125,987/\$6,645 = 18.96$

\$ millions	INVT
Procter & Gamble	$\$42,391/\$7,050 = 6.01$
McDonald's	$\$6,167/\$113 = 54.58$
Valero Energy	$\$115,719/\$5,285 = 21.90$

\$ millions	PPET
Procter & Gamble	$\$83,680/\$20,835 = 4.02$
McDonald's	$\$27,006/\$22,448 = 1.20$
Valero Energy	$\$125,987/\$23,923 = 5.27$

- c. For all three companies, these ratios reflect differences in their businesses, and the overall AT ratio is related to the three individual ratios as seen in Exhibit 5.8 in the chapter. Valero has the highest AT: it collects from its customers very quickly and carries small amounts of inventory relative to its cost of goods sold. It also has the lowest level of property, plant and equipment relative to its sales. Procter & Gamble's ratios are influenced by the relative strength of its largest customer (Wal-Mart), resulting in higher inventory levels and slower collections. In addition, P&G has a large level of intangible assets, as we will see in Chapter 8, so its PPET is relatively high, but its AT is the lowest of the three. McDonald's inventory turnover is very quick because of its perishable nature, and its receivable turnover is higher because most customers pay in cash. The receivables result from the payments due from franchisees.

C12-48 (a) (~ 10 min)

a.

Raising prices or reducing costs **does not always guarantee** an increase in gross profit. To understand why, let's look at both strategies and see how they relate to basic economics concepts, like **supply and demand** and **elasticity**.

Raising Prices

Potential Benefits:

Raising prices means more revenue per unit sold, which can increase gross profit—if customers continue to buy the product at the same rate.

Risks and Limitations:

- ☐ **Customer Reaction:** When prices go up, customers might buy less, especially if there are cheaper alternatives. This can lead to fewer sales, and the total gross profit might end up **lower** than before.
- ☐ **Elasticity of Demand:** If the product is **elastic** (customers are very sensitive to price changes), a small increase in price can lead to a big drop in sales. On the other hand, if the product is **inelastic** (customers still buy even if the price rises), raising prices can help increase gross profit.

Economics Link:

As prices go up, the **quantity demanded** usually goes down. If demand is elastic, the drop in sales reduce total profit instead of increasing it.

Reducing Costs

Potential Benefits:

Reducing the **cost of goods sold (COGS)** lowers expenses and can directly increase gross profit.

Risks and Limitations:

- ☐ **Impact on Product Quality:** If cutting costs means using cheaper materials or less skilled labor, **product quality** might suffer. If customers notice this drop in quality, they may stop buying, which will hurt sales and overall profit.
- ☐ **Short-Term vs. Long-Term Effects:** Reducing costs too much can save money in the short term, but it might hurt production efficiency or brand reputation in the long run, leading to fewer sales.

Economics Link:

Lowering production costs might help increase supply. However, if quality drops, **demand** may shift to the left, meaning fewer customers want to buy, which reduces gross profit.

Conclusion

While raising prices or reducing costs **can** lead to an increase in gross profit, it doesn't always work that way. Whether gross profit increases depends on how **customers respond** to price changes and how **quality** impacts demand. Understanding **supply and demand**, **elasticity**, and **customer expectations** is essential to making decisions that truly benefit the company's bottom line.

C12-49 (~ 10 min)

a.

To improve operating efficiency and the cash conversion cycle, management needs to focus on three key areas: **decreasing inventories**, **decreasing receivables**, and **increasing payables**. Let's look at how each can be done, and then we'll connect them to the overall cash conversion cycle.

1. Decrease Inventories

Decreasing inventories means holding onto less items before they are sold. This is important because inventory sitting on shelves ties up cash, which could be used elsewhere in the business.

Ways to Decrease Inventories:

- ☐ **Order Smaller Quantities More Frequently:** Instead of buying large amounts of inventory, order smaller quantities as needed to keep inventory levels lower. This is often referred to as "just-in-time" inventory.
- ☐ **Improve Inventory Management:** Use better forecasting to understand what customers want and avoid overstocking products that don't sell quickly.
- ☐ **Clear Out Slow-Moving Items:** Offer discounts or promotions to quickly sell items that aren't moving.

By holding less inventory, the business **reduces the amount of cash tied up**, making cash more readily available for other needs.

2. Decrease Receivables

Receivables are the payments that customers owe to the business. Decreasing receivables means **collecting money from customers faster**. When customers take a long time to pay, it ties up cash that could be used in other parts of the business.

Ways to Decrease Receivables:

- ☐ **Shorten Credit Terms:** Reduce the time customers have to pay their bills. For example, instead of allowing 60 days, ask for payment in 30 days.
- ☐ **Encourage Early Payments:** Offer discounts to customers if they pay earlier, such as a 2% discount if they pay within 10 days.
- ☐ **Set Up Effective Collection Processes:** Use reminders and follow-up calls to ensure customers pay on time.

Collecting payments faster means that the business has more cash available sooner, which improves its ability to meet other obligations.

3. Increase Payables

Payables are the amounts the business owes to its suppliers. Increasing payables means **taking more time to pay suppliers**, without risking late fees or damaging relationships.

Ways to Increase Payables:

- ☐ **Negotiate Longer Payment Terms:** Ask suppliers for more time to pay, for example extending terms from 30 days to 60 days.
- ☐ **Use Trade Credit:** Use the full payment period allowed by suppliers, but make sure payments are still on time to maintain a good relationship.

Taking longer to pay suppliers means that the business keeps its cash for a longer period, which can be used for other operational needs.

Connecting to the Cash Conversion Cycle

The **cash conversion cycle (CCC)** measures how long it takes for a company to turn its investment in inventory into cash from sales. It includes the time to:

- ☐ Sell inventory (inventory turnover and days in inventory).
- ☐ Collect cash from customers (receivables turnover and days in receivables).
- ☐ Pay suppliers (payables turnover and days in payables).

By **decreasing inventories and receivables** and **increasing payables**, a business can improve its cash conversion cycle:

Why Is This Important?

Improving the cash conversion cycle means that a company **spends less time with cash tied up** in operations and **more time with cash on hand**. This is crucial for:

- ☐ **Meeting day-to-day expenditures** without needing extra loans.
- ☐ **Investing in new opportunities** more easily.
- ☐ **Improving overall financial health** by having enough cash flow to support business needs.

b.

Reducing receivables and inventories, as well as delaying payments to suppliers, can have significant implications for customers and suppliers. To achieve these goals effectively **without harming performance**, companies need to balance financial efficiency with maintaining positive relationships and customer satisfaction.

Customer Implications of Reducing Receivables and Inventories

1. Reducing Receivables (Collecting Payments Faster)

- **Implication:** Customers may feel pressured if the payment terms are tightened too much. For instance, shortening payment terms or aggressively pursuing payments could lead to a negative customer experience, potentially reducing customer loyalty and future sales.
- **How to Achieve Without Negative Impact:**
 - **Offer Early Payment Incentives:** Encourage customers to pay sooner by offering discounts or other incentives for early payments. This maintains goodwill while improving cash flow.
 - **Communicate Clearly:** When changing payment terms, communicate well in advance, explaining the reason and value behind it. Customers are more likely to respond positively if they understand the change and benefit from rewards like discounts.
 - **Flexible Terms for Key Customers:** Maintain flexible payment terms for large or loyal customers to avoid losing important business relationships. Customizing terms can help balance the need for cash flow with customer satisfaction.

2. Reducing Inventories

- **Implication:** If inventory levels are reduced too much, there's a risk of not being able to meet customer demand, leading to missed sales opportunities and customer dissatisfaction. Customers who can't find what they want may switch to competitors.
- **How to Achieve Without Negative Impact:**
 - **Optimize Inventory Management:** Use technology for **better demand forecasting** to align inventory levels closely with customer needs. This ensures that popular items are always in stock while reducing excess inventory.
 - **Implement Just-in-Time (JIT) Inventory:** Develop efficient supply chain processes that allow inventory to be restocked as needed. This keeps inventory levels low without running out of popular products.
 - **Enhance Supplier Relationships:** Build strong relationships with reliable suppliers who can provide quick replenishment when needed, reducing the risk of stockouts.

Supplier Implications of Delaying Payments

1. Delaying Payments (Extending Payables)

- **Implication:** Suppliers may be unhappy if payments are consistently delayed or if the company demands extended payment terms. This can lead to tension in the relationship, possible disruption in the supply chain, or worse terms from suppliers, like increased prices or reduced priority.
- **How to Achieve Without Negative Impact:**
 - **Negotiate Mutually Beneficial Terms:** Work with suppliers to negotiate extended terms in exchange for a benefit to them, such as a promise of larger order volumes or longer-term contracts. This shows commitment and gives suppliers an incentive to agree to better payment terms.
 - **Transparent Communication:** Maintain open communication with suppliers about payment timelines. Clear, honest discussions can prevent misunderstandings and help suppliers plan accordingly.
 - **Vendor Diversification:** Build relationships with multiple suppliers to reduce dependency on any one vendor. This can provide some leverage in negotiating terms without risking the supply chain if one supplier becomes unhappy.
 - **Pay Strategically:** Prioritize paying strategic suppliers on time or even early, while extending payments with others. This ensures that key partners are kept happy, maintaining a smooth supply chain while achieving cash flow improvements.

Conclusion:

Reducing receivables and inventories and extending payables can improve cash flow, but it needs careful handling to avoid hurting relationships with customers and suppliers. Using a balanced approach, focusing on communication, incentives, and technology, businesses can meet these financial goals while still maintaining strong operational performance and stakeholder satisfaction.